

The Interrogation — Session 4 Lab Sheet

Saturday 5 September · your Part C output IS your P3, committed before 5:00 PM

Name: Sanchez, Eliah Therrien E. **Track:** P5 · _____

Ground rules: every chart you make must be *read* — a Markdown sentence under it saying what it shows and what it means for prediction. This session forward, **reading cells are graded**. Every predictor gets the calendar test: *knowable on the morning you predict?* Datasets load from Drive, never get committed. Nothing changes silently — print your counts.

PART A · GUIDED — YOUR SLICE, ON THE TABLE

Interrogate your own table (2:30–3:15)

Work on YOUR P2 slice. *No P2 slice yet? Build it now with the quick-start below, then continue — your P2 package is still owed separately.*

```
# QUICK-START (only if you have no slice yet): branch-month revenue
import pandas as pd
URL = "https://drive.google.com/uc?export=download&id="
sales = pd.read_csv(URL + "1LlA5ni-10NzZq6ODU4eq8A1B6zi0WlDA")
sales = sales.drop_duplicates()
sales["BranchID"] = (sales["BranchID"].str.strip()
                    .str.upper().str.replace("-", ""))
sales["Date"] = pd.to_datetime(sales["Date"], errors="coerce")
sales = sales.dropna(subset=["Date"])
sales["YM"] = sales["Date"].dt.to_period("M").astype(str)
df = sales.groupby(["BranchID", "YM"], as_index=False)["Amount"].sum()
```

A1 Reload & re-handshake · 5 min

1. Load your slice, run `info()` and your row count. Rows: **108** Does it match your P2? **Yes**. If not, stop and find out why before anything else.

A2 Univariate — is the average honest? · 12 min

```
df["<measure>"].describe()
df["<measure>"].skew()
df["<measure>"].hist(bins=30)
```

2. Mean: **23246.720370** Median: **23005** Skewness: **np.float64(0.19533492889659224)**

3. One sentence: does the mean fairly represent a typical row of your slice — and what would a model trained to “predict the average” get wrong?

- The mean fairly represents a typical row because it is close to the median, but a model that only predicts the average might not show the differences between low and high revenue.

A3 The rhythm — what repeats? · 13 min

Plot your measure over time (line per branch/product, or monthly bars). Hunt for: trend up/down · seasonality · a break in the pattern.

4. The strongest repeating pattern in your slice, WITH a number attached (e.g., “December runs ~25% above the monthly average”):

- The strongest pattern is that the highest month is about 30.4% higher than the monthly average.

A4 The outlier hunt — dirt or loud truth? · 15 min

```
q1, q3 = df["<measure>"].quantile([.25, .75])
fence_hi = q3 + 1.5*(q3 - q1)
df[df["<measure>"] > fence_hi]
```

5. Rows beyond the fence: **1**. Pick your most extreme row and rule on it — KEEP / FLAG / DROP — with the reason a reader would accept:

- The most extreme row is B01 in 2025-12 with an amount of 44,885. I would KEEP it because it could be a real high-revenue month and there is no proof that it is wrong.

Remember the demo: the fence is a flashlight, not a judge. An outlier that is real is signal about how big days happen — deleting it teaches your model that they don't.

PART B · HUNT THE LEAK

Ten columns walk in. Five are lying. (3:25–3:50)

The scenario: it is the FIRST MORNING of the month. The Operations lead must predict this month's revenue per branch (MonthRevenue_TARGET) to set staffing and stock. The file `hunt_the_leak.csv` (FILE_ID posted in the LMS) offers ten candidate predictor columns. **Five of them cheat.**

Your weapon is ONE question, applied to every column: **“Could Kape Tayo actually know this number on the morning of day 1?”** Correlation may rank candidates — the calendar disqualifies them. (One planted leak has an innocent, near-zero correlation. Correlation will not save you.)

6. Complete the audit — verdict and one-line reason for every column:

Column	Knowable day 1?	Verdict	Why (one line)
MonthNo	Yes	KEEP	The month number is known before the month starts.
PromoDaysPlanned	Yes	KEEP	Planned promo days can be known before the month starts.
StaffHoursScheduled	Yes	KEEP	Scheduled staff hours can be planned before the month starts.
PrevMonthOrders	Yes	KEEP	Last month's orders are already known.
PrevMonthRevenue	Yes	KEEP	Last month's revenue is already known.
OrderCount (this month)	No	DROP	This month's orders are only known after the month starts.
AvgTicket (this month)	No	DROP	It uses this month's sales information, which is not known on Day 1.
MonthSalesTotal	No	DROP	This month's total sales are only known after sales happen.
RefundsProcessed	No	DROP	Refunds processed during the month are future information.
QuarterRevenueTotal	No	DROP	It includes revenue from the current/future quarter that is not known on Day 1.

7. Name the SNEAKIEST leak you found — the one most likely to survive into a careless model — and say what makes it sneaky:

- The sneakiest leak is QuarterRevenueTotal because it looks like a normal past revenue value, but it can include information from the current or future months. It is easy for a model to use it without noticing that it is not available on Day 1.

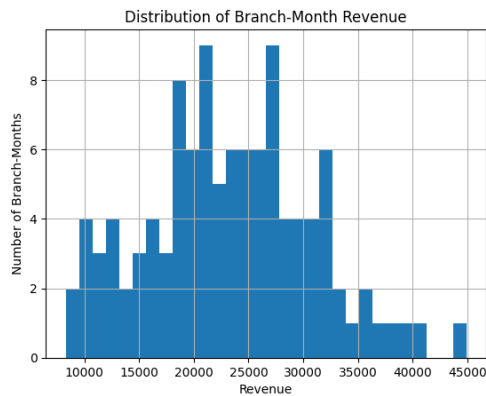
PART C · INDIVIDUAL — YOUR P3

The EDA package (3:50–4:40)

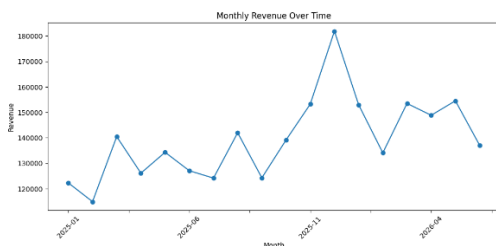
P3 = an EDA notebook on YOUR slice with four deliverables. The Discussion → Code → Reading rhythm is the graded format — a notebook of bare code caps at Developing regardless of what the code does.

C1 Three annotated charts (minimum) · 20 min

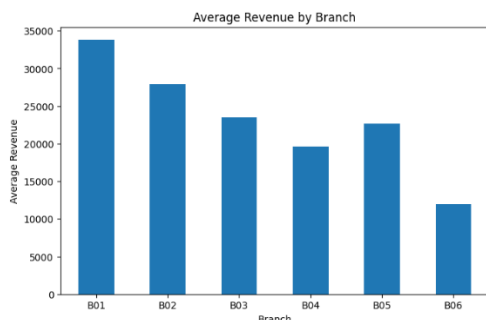
Each chart earns its place: titled, axes labeled with units, and a **reading cell under it** answering two questions — what does it show, and what does it mean for PREDICTION? Suggested trio: your A2 distribution · your A3 rhythm · one comparison of your choosing (branch vs branch, product vs product, weekday vs weekend).



The revenue is mostly around the middle, with some higher values. For prediction, this means the model should not only expect the average revenue because some months are higher than others.



The revenue changes between months, and the highest month is about 30% above the monthly average. For prediction, the month can matter because revenue is not the same every month.



The branches have different average revenue levels. For prediction, BranchID could be useful because some branches may normally earn more than others.

C2 Leakage audit — on your own predictors · 10 min

List every candidate predictor your P1 field 7 named (plus any new ones EDA suggested), and run the calendar test on each — same table as Part B, in your notebook:

My candidate predictor	Knowable in advance?	Verdict	Why
MonthNo	Yes	KEEP	The month number is known before the month starts.
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PrevMonthOrders	Yes	KEEP	Last month's orders are already known.
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C3 Base rate & balance note • 8 min

Compute YOUR track's base rate and write one sentence on what it means for your future model:

Track	Your base rate to compute	Why it matters
P4 • MKT	Weekend share of total revenue (%)	The naive "weekends matter" floor
P5 • FIN	Revenue share of the top 2 products (%)	Concentration — a 12-product forecast may really be a 2-product forecast
P6 • HR	Smallest branch headcount share (%)	Small groups = unstable rates — the imbalance lesson at branch grain
P7 • OPS	Quietest ÷ busiest branch-month order ratio	The spread a single flat forecast would smear over

8. My base rate: **28.98%** Implication, one sentence: This means the revenue is not only dependent on two products, so the model should consider more than just the top products when making a forecast.

C4 The predictive-signal memo • 10 min

Five sentences, in your notebook, addressed to your P1 reader — fill each slot with a NUMBER, not an adjective:

Slot	Your sentence
1 • Strongest signal found	"The clearest predictable pattern is ... (magnitude: ...)"
2 • Second signal	"There is also ... (magnitude: ...)"
3 • One trap found & disqualified	"I considered ... but it fails the calendar test because ..."
4 • The floor to beat	"Any model must beat ... (my baseline / base rate: ...)"
5 • Modeling hypothesis	"Week 6, I will first try predicting ... using ..."

C5 Package and commit • 7 min + the commit block

P3 checklist — in <your-folder>/p3-eda/: ☐ notebook (Restart & Run all clean, reading cells throughout) ☐ ≥3 annotated charts ☐ leakage audit table ☐ base-rate note ☐ signal memo ☐ AI-use log entry WITH conversation link. Commit: "P3: <surname> – EDA, leakage audit, signal memo v1". Content freezes 5:00 PM; copy-editing until 11:59 tonight.

Next Saturday is the PRELIM (Rubric A): your P2 + P3, defended in a 10-minute oral — every decision, without notes. The work you commit today is the work you will stand behind next week. Build it like the panel is already listening.